How-To Library

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Secure@Source Data Store Configuration Reference

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Abstract

This article lists the connectivity steps and permissions required for Secure@Source to access and scan data stores.

Supported Versions

- Secure@Source 4.5

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Secure@Source Data Store Configuration Overview

Before you create and scan data stores in Secure@Source, you must first configure connectivity and assign permissions for Secure@Source to access the source.

Amazon Redshift

Objects Accessed

- Tables
- Views

Amazon Redshift Permissions

If you create a new user, ensure that you configure read permission on the Amazon Redshift data source for the user account.

Amazon S3

Objects Accessed

File

Permissions

- Configure read permission on the Amazon S3 data source for the user account that you use to access the data source.
- Configure access permission for the user account if you use a user account that is different from the user account used to create the Amazon S3 data source.

Azure Data Lake Store

Objects Accessed

File

Permissions

If you create a new user, ensure that you configure read permission on the data source for the new user account.

Azure Microsoft SQL

Objects Accessed

- Tables
- Views
- Extended stored procedure
- Table (user-defined)
- Synonym
- UNIQUE constraint
- Triggers
- Procedures
- Index
- SQL table-valued-function
- Users
- Partition scheme
- Partition function
- XML schema collections
- User-defined data type
- Rules
- Defaults
- Constraints
- SQL_INLINE_TABLE_VALUED_FUNCTION
- SQL_SCALAR_FUNCTION
- SEQUENCE

Azure Microsoft SQL Permissions

Before you create an Azure Microsoft SQL data store, configure the permissions for the Azure Microsoft SQL Server database user account that you use to connect to Azure Microsoft SQL Server.

Secure@Source uses Microsoft SQL Server authentication to connect to the Azure Microsoft SQL Server database. The user account that you use to connect to Azure Microsoft SQL Server must be a Microsoft SQL Server login account.

Configure the VIEW DEFINITION permission for the user account. Configuring this permission lists all the schemas from where the user can load metadata.

Azure WASB

Objects Accessed

File

Permissions

Configure read permission on the Azure WASB data source for the user account that you use to access the data source.

Cloudera Navigator

Objects Accessed

- HDFS files, directories, and datasets
- Hive tables, query templates, and query executions
- Impala query templates and executions
- MapReduce job templates and executions
- Oozie job templates and executions
- Pig tables, job templates, and job executions
- Sqoop job templates and executions
- YARN job templates and executions

Permissions

Heap Size

Before you create a Cloudera Navigator data store, configure the Java heap size for the Cloudera Navigator server and the maximum heap size for Catalog Service. If you do not correctly configure the heap sizes, the metadata load can fail.

Configure the following heap sizes:

- Set the Java heap size for the Cloudera Navigator Server to at least 2 GB. If the heap size is not sufficient, the resource load fails with a connection refused error.
- Set the maximum heap size for the Catalog Service to at least 4096 MB (4 GB).

If you perform simultaneous resource loads, increase the maximum heap size by at least 1024 MB (1 GB) for each resource load. For example, to load two Cloudera Navigator resources simultaneously, increase the maximum heap size by 2048 MB (2 GB). Therefore, you would set Max Heap Size to at least 6144 MB.

Note: Some Cloudera distributions might require a maximum heap size larger than 4 GB. If the metadata load fails with an out of memory error, increase the maximum heap size.

Cloudera Manager User Account

The Cloudera Navigator scan job requires a Cloudera Manager user account that includes the Full Administrator or Navigator Administrator role. The roles include authorization to connect to and retrieve metadata from Cloudera Navigator. If the user account does not have one of these roles, the scan job fails.

File System

Objects Accessed

File

Permissions

Configure read permission on the file system data source for the user account that you use to access the data source.

HDFS

Objects Accessed

File

Permissions

Configure read permission on the HDFS data source for the user account that you use to access the data source.

Hive

Objects Accessed

- Table
- View

Hive Permissions

Configure read permission on the Hive data source for the user account that you use to access the data source.

IBM Db2

Objects Accessed

- Tables
- Partitioned table
- Range clustered table

- Multidimensional clustering table
- Constraints (NOT NULL , UNIQUE , CHECK , PRIMARY KEY , FOREIGN KEY , DEFAULT)
- Alias
- Sequences
- Views
- Triggers
- Function
- Procedure
- Index
- Materialized query table

IBM Db2 Permissions

The database user account must have CREATEIN, DROPIN, and ALTERIN privileges on the NULLID schema. Optionally, you can grant the BINDADD privilege to the user account.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the IBM Db2 database user account:

Schema Objects	System Tables
- Tables - Views - Indexes - Packages - Procedures - Functions - Sequences - Triggers - Synonyms	- SYSCAT.SCHEMATA - SYSCAT.TRIGGERS - SYSCAT.INDEXES - SYSCAT.INDEXCOLUSE - SYSCAT.REFERENCES - SYSCAT.TABCONST - SYSCAT.KEYCOLUSE - SYSCAT.COLCHECKS - SYSCAT.DATATYPES - SYSCAT.COLUMNS - SYSCAT.TABLES - SYSCAT.FUNCTIONS - SYSCAT.PROCEDURES - SYSCAT.PACKAGES - SYSCAT.TRIGGERS - SYSCAT.SEQUENCES - SYSCAT.INDEXES - SYSCAT.CHECKS - SYSCAT.VIEWS - SYSCAT.PACKAGES - SYSCAT.FUNCDEP - SYSCAT.TRIGDEP - SYSCAT.VIEWDEP

IBM Db2 for z/OS

Objects Accessed

- Store procedure
- Function

- Synonym
- Trigger

IBM Db2 for z/OS Permissions

When you load an IBM Db2 for z/OS resource, Secure@Source accesses metadata from the IBM Db2 for z/OS subsystem. Secure@Source selects the metadata from the IBM Db2 catalog tables.

Configure SELECT privileges on the following system tables in the IBM Db2 for z/OS user account that you use to connect to the IBM Db2 for z/OS data source:

System Tables	Metadata Extracted
- SYSIBM.SYSSYNONYMS - SYSIBM.SYSTABLES - SYSIBM.SYSROUTINES - SYSIBM.SYSSEQUENCES - SYSIBM.SYSDATATYPES	Schemas
- SYSIBM.SYSTABLES - SYSIBM.SYSTABLESPACE - SYSIBM.SYSVIEWS	- Tables - Views - Alias
- SYSIBM.SYSTABLES - SYSIBM.SYSCOLUMNS - SYSIBM.SYSTABLESPACE	Columns
- SYSIBM.SYSKEYS - SYSIBM.SYSINDEXES	Primary keys
- SYSIBM.SYSRELS - SYSIBM.SYSFOREIGNKEYS - SYSIBM.SYSINDEXES	Foreign keys
- SYSIBM.SYSCHECKS - SYSIBM.SYSCHECKDEP	Check constraints
- SYSIBM.SYSINDEXES - SYSIBM.SYSKEYS	Indexes
- SYSIBM.SYSSYNONYMS - SYSIBM.SYSTABLES	Synonyms
SYSIBM.SYSTRIGGERS	Triggers
SYSIBM.SYSROUTINES	Stored procedures
SYSIBM.SYSROUTINES	Functions
SYSIBM.SYSSEQUENCES	Sequences
- SYSIBM.SYSSEQUENCESDEP - SYSIBM.SYSSEQUENCES	Identity
SYSIBM.SYSDATATYPES	User-defined data types

Note: If the PowerExchange Listener does not use IBM Db2 security, configure SELECT privileges for the PowerExchange Listener user ID instead of the IBM Db2 for z/OS user account.

Informatica Axon

Objects Accessed

Glossary

Permissions

None.

Informatica PowerCenter

Objects Accessed

Mapping

Permissions

- Configure the Access Repository Manager privilege for the user who accesses the PowerCenter repository.
- Configure read permission on the PowerCenter data source for the user account that you use to access the PowerCenter data source.
- Run the `pmrep ObjectExport` command to export the mappings in PowerCenter.

JDBC with MySQL

Objects Accessed

- Tables
- Constraints (NOT NULL , UNIQUE , CHECK , PRIMARY KEY , FOREIGN KEY , DEFAULT)
- Clone table
- Indexes
- Views
- Triggers
- Function
- Procedure
- Materialized view
- MV trigger

Permissions

You must configure the CONNECT and SELECT_CATALOG_ROLE permissions for the database user account that you use to access the MySQL database.

JDBC with AS400

Objects Accessed

- Tables
- Constraints (NOT NULL , UNIQUE , CHECK , PRIMARY KEY , FOREIGN KEY , DEFAULT)
- Alias
- Duplicate table
- Sequences
- Views
- Triggers
- Function
- Procedure
- Index
- User-defined data types

Permissions

You must configure the CONNECT and SELECT_CATALOG_ROLE permissions for the database user account that you use to access the AS400 database.

Microsoft Azure Blob Storage

Objects Accessed

File

Permissions

Configure read permission on the Microsoft Azure Blob Storage data source for the user account that you use to access the data source.

Microsoft OneDrive

Objects Accessed

File

Permissions

Configure read permission on the Microsoft OneDrive data source for the user account that you use to access the data source.

Microsoft SharePoint

Objects Accessed

File

Permissions

Configure read permission on the Microsoft SharePoint data source for the user account that you use to access the data source.

Microsoft SQL Server

Objects Accessed

- Tables
- Views
- Extended stored procedure
- Table (user-defined)
- Synonym
- UNIQUE constraint
- Triggers
- Procedures
- Index
- SQL table-valued-function
- Users
- Partition scheme
- Partition function
- XML schema collections
- User-defined data type
- Rules
- Defaults
- Constraints
- SQL_INLINE_TABLE_VALUED_FUNCTION
- SQL_SCALAR_FUNCTION
- SEQUENCE

Permissions

Before you create a Microsoft SQL Server data store, configure the permissions for the Microsoft SQL Server database user account that you use to connect to Microsoft SQL Server.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the Microsoft SQL Server database user account:

Schema Objects	System Tables
Tables Views Indexes Packages Procedures Functions Sequences Triggers Synonyms	sysobjects sysusers master.dbo.sysservers syscolumns sysindexkeys systypes syscomments sysreferences sysdepends sys.synonyms (Microsoft SQL Server 2005) sysindexes sys.sql_expression_dependencies

In addition, the user must have VIEW DEFINITION permission on the database.

Netezza

Objects Accessed

- Column
- Data type
- Distribution key
- Distribution key column
- Expression
- External table
- Foreign key
- Materialized view
- Primary key
- Schema
- Sequence
- Synonym
- Table
- Unique key constraint
- View

Permissions

Configure the LIST privilege for the Netezza user to extract metadata from a Netezza database. The LIST privilege grants the Netezza user access to all database objects in the Netezza database through Secure@Source. If required, you can assign database permissions to limit access to individual database objects.

Configure SELECT permissions for the Netezza user account on the following system views:

- _V_TABLE_DIST_MAP
- _V_TABLE
- _V_DATABASE
- _v_relation_keydata
- _v_relation_column
- _v_view
- _v_synonym
- _v_datatype
- _v_sequence

Oracle

Objects Accessed

- Table
- Synonym
- View
- Different datatypes
- Precision and scale information for all data types
- Comments
- Trigger
- Function
- Procedure
- Condition
- Integrity constraints
- Index-organized table
- Materialized view

Permissions

You must configure the CONNECT and SELECT_CATALOG_ROLE permissions for the database user account that you use to access the Oracle database.

If you cannot configure SELECT_CATALOG_ROLE permission for the user account, configure the following permissions:

- SELECT permissions on the v$version and v$instance views.
- CONNECT and SELECT permissions on the required schema objects and corresponding system tables.

Configure SELECT permissions on the following schema objects:

- Tables
- Views
- Indexes
- Packages
- Procedures
- Functions
- Sequences
- Triggers
- Synonyms

Configure SELECT permissions on the following system tables:

- ALL_OBJECTS
- ALL_SOURCE
- ALL_USERS
- DBA_ARGUMENTS
- DBA_COL_COMMENTS
- DBA_CONS_COLUMNS
- DBA_CONSTRAINTS
- DBA_DEPENDENCIES
- DBA_IND_COLUMNS
- DBA_INDEXES
- DBA_MVIEWS
- DBA_OBJECTS
- DBA_PROCEDURES
- DBA_RECYCLEBIN
- DBA_SEQUENCES
- DBA_SOURCE
- DBA_SYNONYMS
- DBA_TAB_COLUMNS
- DBA_TAB_COMMENTS
- DBA_TAB_PARTITIONS
- DBA_TABLES
- DBA_TRIGGERS
- DBA_VIEWS
- SYS.V_$DATABASE
- SYS.DBA_USERS
- SYS.DBA_DIRECTORIES
- SYS.DBA_EXTERNAL_LOCATIONS
- SYS.DBA_CLU_COLUMNS

If schemas have a table with a cluster defined on some columns, you must configure SELECT permission on the DBA_CLUSTERS system table.

Salesforce

Objects Accessed

Table

Permissions

Before you import user metadata from Salesforce, verify that the user account has the required privileges:

- The Salesforce org has the Bulk API feature enabled.
- The user account has the View All Data and the View Event Log Files privileges.

SAP

Objects Accessed

- Application components
- Packages
- Tables
- Columns
- Primary and foreign keys

Permissions

The scan SAP data stores, the user account must have the following authorizations:

Authorization Object	Activity	Purpose
S_DEVELOP	All activities.	Install and uninstall programs.
S_TRANSPRT	CREATE and CHANGE	Install and uninstall programs. Required for workbench requests TTYPE = DTRA.
S_TABU_DIS	READ	Extract data.
S_DATASET	READ and WRITE	Read/Write staging data to a file.

Authorization Object	Activity	Purpose
S_BATCH_JOB	DELE, LIST, PLAN, SHOW	Release background jobs.
S_RFC	All activities for RFC objects in the following function groups: - ZPMV - ZERP - ZPMH - ZPMR - ZPMD - ZPMI - ZPMF - SYST - RFC1 - /INFATRAN/* - SDIFRUNTIME - GSAC - SKEY - STR9 - SEU_COMPONENT - STRD - BTCH - SABC - KXUT - /1BCDWBEN/SEN4 - EDI1	Authorize RFC privileges.

Sybase

Objects Accessed

- Table
- View
- Clustered index
- Procedure
- Trigger
- Rule
- Unique index
- Non-clustered index
- Index
- Function
- User-defined data type

Permissions

Before you create a Sybase resource, configure the permissions for the Sybase ASE database user account that you use to connect to the Sybase ASE database.

Configure SELECT permissions on the following schema objects:

- Tables
- Views
- Columns
- Procedures

Configure SELECT permissions on the following system tables:

- sysobjects
- syscomments
- sysdepends
- syscolumns
- sysindexes
- sysconstraints
- sysreferences
- sysusers
- systypes
- master.dbo.sysservers
- master.dbo.sysconfigures
- @@<servername>

Teradata

Objects Accessed

- User
- Table
- Column
- Datatype
- View
- Index
- IndexColumn
- PrimaryKeyConstraint
- ForeignKey
- UniqueConstraint
- CheckConstraint
- Procedure
- TableTrigger
- Macro

Permissions

Configure the permissions for the Teradata database user account that you use to connect to the Teradata database.

The following table lists the schema objects and system tables on which you must configure SELECT permissions for the Teradata database user account:

Schema Objects	System Tables
Tables Views Indexes Packages Procedures Functions Sequences Triggers Synonyms	DBC.ALL_RI_CHILDREN DBC.COLUMNS DBC.DATABASES DBC.DBASE DBC.DBCINFO DBC.HOSTSINFO DBC.INDICES DBC.SHOWTBLCHECKS DBC.TABLES DBC.TABLETEXT DBC.TRIGGERS DBC.TVFIELDS DBC.TVM DBC.UDTINFO TDQCD.STATSRECS

Author